

THREE LOCAL INVERSE POINTS, TWO RAMIFICATION TYPES, AND COMPLETE FIXED-COFACTOR ERDŐS–STRAUS RETURNS

THE CLANKERS

ABSTRACT. The reciprocal cubic of an original prime Erdős–Straus state is a literal target of the four-dimensional ES–Fabel polynomial. Its seven-point affine fibre always has exactly three rational points over the original local field. Exterior states give two ramified quadratic factors; middle states give two unramified quadratic factors. The original prime character distinguishes the algebras although rational-point counts do not. An explicit negative-square shape transfer changes that arithmetic target at the same prime and returns a middle state. The full fixed-cofactor divisor fibre is then classified and proved bijective with all original middle states having that cofactor. Universal existence is not asserted.

1. ORIGINAL COORDINATES

Fix a prime $p \equiv 1 \pmod{4}$, $p/4 < a < p/2$, $R = 4a - p$, and $u \mid a^2$. The original gates are $E : R \mid 4u + 1$ and $M : R \mid u + a$. Set $g = \gcd(a, u)$, $(h, r, s) = (g^2/u, u/g, a/g)$. For E let $\kappa = (pr + s)/R$ and $(x_1, x_2, x_3) = (a, h\kappa, phr\kappa)$; for M let $\lambda = (r + s)/R$ and $(x_1, x_2, x_3) = (a, phs\lambda, phr\lambda)$. These are the classical Type-I/II coordinates [1, §2]. They give $4/p = \sum_i 1/x_i$, and their ordered inverses are $u = a^2/(Rx_2 - pa)$ and $u = pa^2/(Rx_2 - pa)$.

Lemma 1. *Every original square divisor has $(u/R) = (u/p)$. In particular E implies $(h/p) = -1$, and M implies $(rs/p) = -1$.*

Proof. For odd $l \mid a$, reciprocity gives $(l/R) = (-1/l)(R/l) = (p/l) = (l/p)$. For $2 \mid a$, use $R \equiv -p \pmod{8}$ and the supplementary law. Multiply over the original exponents. The E target is $-1/4$; for M use $r \equiv -s \pmod{R}$ and $(-1/R) = -1$. $\square$

The ordered reciprocal roots $t_i = p/x_i$ are distinct and have sum four. Retain the cubic and its quartic target

$$f(T) = \prod_i (T - t_i) = T^3 - 4T^2 + \sigma_2 T - \sigma_3, \quad H(U, V) = V \prod_i (U - t_i V).$$

The coefficient map to the existing ES–Fabel polynomial is $\mathbf{U} = (0, -4, \sigma_2, -\sigma_3)$. Its full marked arithmetic inverse is $x_i = p/t_i$, subject to positive integrality, followed by the original divisor inverse above. Neither p nor the ordering is discarded.

2. THE ORIGINAL POLYNOMIAL FIBRE

For the exact four-dimensional polynomial P in [2, GF1], its factor coordinates have $L = \xi U + bV$ and $C = cU^3 + dU^2V + eUV^2 + f_0V^3$, with $\xi d + bc = 1$ and $C(-b, \xi) = 1$; their product is $P_0U^4 + U^3V + P_2U^2V^2 + P_3UV^3 + P_4V^4$. For $d_i = f'(t_i)$, the full finite inverse is

$$\begin{aligned} \xi_i^2 &= d_i^{-1}, \quad y_i = -t_i - i/\xi_i, \quad z_i = 2t_i/\xi_i + 3i/\xi_i^2, \\ w_i &= 7it_i^2/\xi_i + (-4 - 17t_i)/\xi_i^2 - 13i/\xi_i^3. \end{aligned} \tag{1}$$

Both signs occur. This follows by factoring at $t_i = -b/\xi_i$ and evaluating $C(-b, \xi_i) = \xi_i^2 f'(t_i)$. The factor-coordinate inverse in [2, GF5–7] gives (1). The single infinity chart is

$$(0, -4, i(112 - \sigma_2)/2, -704 + 8\sigma_2 + \sigma_3). \tag{2}$$

The opposite projective factor sign at infinity is absent from this particular affine chart. Thus there are seven geometric points. The polynomial Jacobian is identically -2 ; its original constant is retained.

Fix either embedding $i \mapsto i_p \in \mathbb{Q}_p$. The complete fibre algebra is

$$\mathcal{A} = \mathbb{Q}_p \times \prod_{i=1}^3 \mathbb{Q}_p[\xi_i]/(\xi_i^2 - d_i^{-1}). \quad (3)$$

Theorem 2. *Every original E or M state has exactly three $\mathbb{Q}_p$ -rational inverse points. Its complete algebra is*

$$E: \mathbb{Q}_p^3 \times K_{\text{ram}}^2, \quad M: \mathbb{Q}_p^3 \times K_{\text{unr}}^2,$$

with respectively a ramified and the unramified quadratic extension. The two copies retain different original root labels.

Proof. For E let $H = hrs\kappa$. The roots are $(p\kappa, pr, s)/H$, with H a p -unit. The difference $\kappa - r$ is also a unit: otherwise $pD + 1 = 4hr\kappa$ would make the nonresidue hr^2 equal to $1/4 \pmod{p}$. Consequently the derivative valuations are $(1, 1, 0)$, $d_1/d_2 \equiv -1 \pmod{p}$ and $d_3 \equiv 16 \pmod{p}$. Unit square residues lift to squares, so the first two factors are copies of one ramified quadratic field and the last splits.

For M let $H = hrs\lambda$. The roots are $(p\lambda, r, s)/H$, with distinct reductions. Up to H^{-2} , their derivatives reduce to $rs, r(r-s), s(s-r)$. The first is a nonsquare by the lemma. Their product is the square $-r^2s^2(r-s)^2$, so precisely one is a square and two are nonsquare units. The latter define the same unramified quadratic field. Add the single infinity factor in both cases. The standard square-unit lifting step is explicitly $b_{n+1} = b_n + p^n t$, with $t = -(b_n^2 - z)p^{-n}(2b_n)^{-1} \pmod{p}$. $\square$

In E, $\bar{f} = T^2(T-4)$, $v_p(\sigma_2, \sigma_3, \text{Disc } f) = (1, 2, 2)$. Formula (1) gives valuation vector $(-1/2, 1/2, 1, 1)$ for all four branches above the first two roots. They are not integral. Only the two branches at four and the infinity point specialize, leaving three geometric special points. In M all seven branches are integral over unramified extensions, and seven geometric special points remain. Both special fibres nevertheless have three $\mathbb{F}_p$ -rational points. The difference is not a zero source Jacobian: -2 is still a unit.

The original complex conductor conjugacy in [2, FC32–35] transports all seven points and their original norms. It is not thereby a $\mathbb{Q}_p$ -model of its complex period matrix. Here the arithmetic statements apply to the literal polynomial over $\mathbb{Q}(i)$ with the specified embedding.

The integral root order records the lost arithmetic. Evaluation identifies the E root order with $\{z \in \mathbb{Z}_p^3 : z_1 \equiv z_2 \pmod{p}\}$, of index p and conductor $p\mathbb{Z}_p \times p\mathbb{Z}_p \times \mathbb{Z}_p$; the M order is all $\mathbb{Z}_p^3$. Necessity follows from congruence of the first two E roots. Sufficiency follows by integral interpolation at those two roots and a correction by $(T-t_1)(T-t_2)$ at the separated third root.

For any same-prime E and M root triples x_i, y_i , the unique labelled quadratic interpolation $F(x_i) = y_i$ is $F(T) = \sum_i y_i(T-x_j)(T-x_k)/[(x_i-x_j)(x_i-x_k)]$. It has coefficient valuations $(0, -1, -1)$ and $pF \pmod{p}$ is a nonzero multiple of $T(T-4)$. The reverse interpolation is integral. The claims follow directly from the derivative valuations: after multiplying by p , only the second summand survives modulo p . Thus a rational root-level isomorphism and its precise integral obstruction both exist. This root-order conductor is distinct from the weighted complex conductor in the Zeta source.

3. CHANGING THE ARITHMETIC TARGET AT THE SAME PRIME

For E put $v = hs^2$, $D = (4u+1)/R$, $\alpha = v - R$, $\beta = v - D$. Direct reduction gives

$$h \mid \alpha\beta - 1, \quad p \equiv \alpha \pmod{h}, \quad p\beta \equiv 1 \pmod{h}.$$

Theorem 3. *If $\alpha = -4c^2$ or $\beta = -4c^2$ for a positive integer c , then $h \equiv 3 \pmod{4}$ and $\gcd(h, c) = 1$. Put $j = (h+1)/4$. On the exact domain $c \mid j$, set*

$$Q = h, \quad U = c^2 \text{ (}\alpha \text{ case)}, \quad U = (j/c)^2 \text{ (}\beta \text{ case)}, \quad R_M = (p+4U)/h, \quad a_M = (p+R_M)/4.$$

This is an original M state with

$$g_M = \gcd(j, U), \quad (h_M, r_M, s_M, \lambda_M) = (g_M^2/U, U/g_M, R_M j/g_M - U/g_M, j/g_M).$$

In particular every existing original E state on either line $\alpha = -4$ or $\beta = -4$ has a middle return; no assertion that the lines themselves contain infinitely many source states is made.

Proof. In the alpha branch $R = hs^2 + 4c^2 \equiv 3 \pmod{4}$; in the beta branch $D = hs^2 + 4c^2 \equiv 3 \pmod{4}$ follows from $RD = 4hr^2 + 1$ and $R \equiv 3 \pmod{4}$. Thus s is odd and $h = 4j - 1$. The shape congruences also make h coprime to c . In the first case $h \mid p + 4c^2$. In the second, $p \equiv -(4c^2)^{-1}$ and $4j \equiv 1 \pmod{h}$ give $h \mid p + 4(j/c)^2$. The stated divisibility is precisely what puts the chosen word in the actual box $U \mid j^2$. For the first case $4U < p - h$ follows from the original $R = hs^2 + 4c^2 < p$. For the second, $4U \leq (h+1)^2/4 < (h-1)p$, since $p > 2h$ and $h \geq 3$. Thus $0 < R_M < p$. The displayed gcd normalization gives $a_M = h_M r_M s_M$, $U = h_M r_M^2$ and $R_M \lambda_M = r_M + s_M$. Any prime dividing r_M, R_M would divide p , impossible because $r_M \leq j < p$. The normalization is primitive and the ordered M identity follows. With $j = cm$, $d = \gcd(c, m)$, the new grade is explicitly $h_M = d^2$. $\square$

At the hard prime $p = 5209$, the original E state

$$(a, R, u, h, r, s, \kappa) = (1330, 111, 18620, 95, 14, 1, 657)$$

and shape $(-16, -576)$. Its two marked returns ($c = 2$ on α , $c = 12$ on β) coincide at $j = 24, U = 4$. The M state is

$$(a_M, R_M, u_M, h_M, r_M, s_M, \lambda_M) = (1316, 55, 4, 4, 1, 329, 6),$$

giving

$$\frac{4}{5209} = \frac{1}{1316} + \frac{1}{41130264} + \frac{1}{125016}.$$

The actual new factorization contains 47, absent from the old E source. The local fibre changes from ramified to unramified because the target has changed, not through an impossible isomorphism of the unchanged local algebras.

Availability cannot be suppressed. At $p = 3049$, the E state $(806, 175, 20956, 31, 26, 1, 453)$ has $\alpha = -144$, but $c = 6$ does not divide $j = 8$. No divisor of 64 is the required 5 $\pmod{31}$, so the entire selected cofactor box is empty. Another original M shell, $a = 770, u = 5, R = 31$, succeeds.

4. THE COMPLETE FIXED-COFACTOR MIDDLE FIBRE

The selected square words in Theorem 3 sit inside a larger finite fibre which can be computed without guessing a word.

Theorem 4 (Complete fixed- Q middle fibre). *Let*

$$(p, a, R, u, h, r, s, \kappa, D, \alpha, \beta)$$

be an original E state, in the notation of the preceding sections, and assume $h \equiv 3 \pmod{4}$. Put

$$j = \frac{h+1}{4}, \quad \mathcal{U}_{p,h} = \{U \in \mathbb{Z}_{>0} : U \mid j^2, h \mid p + 4U\}.$$

For $U \in \mathcal{U}_{p,h}$, define

$$\begin{aligned} R_U &= \frac{p+4U}{h}, & a_U &= \frac{p+R_U}{4}, & g_U &= \gcd(j, U), \\ h_U &= \frac{g_U^2}{U}, & r_U &= \frac{U}{g_U}, & \lambda_U &= \frac{j}{g_U}, & s_U &= R_U \lambda_U - r_U. \end{aligned} \quad (19)$$

Then

$$\boxed{(a_U, R_U, U, h_U, r_U, s_U, \lambda_U)}$$

is an original M state at the same prime. Its middle cofactor is exactly

$$Q_U = 4h_U r_U \lambda_U - 1 = h,$$

and its ordered reciprocal decomposition is

$$\boxed{(a_U, ph_U s_U \lambda_U, ph_U r_U \lambda_U)}. \quad (20)$$

The assignment $U \mapsto (a_U, R_U, U, h_U, r_U, s_U, \lambda_U)$ is a bijection from $\mathcal{U}_{p,h}$ onto all original M states at p whose normalized cofactor is $Q = h$.

Proof. Because $h \leq a < p/2$, $p > 2h$, and $h \geq 3$. From $U \mid j^2$,

$$4U \leq 4j^2 = \frac{(h+1)^2}{4} < 2h(h-1) < p(h-1).$$

Consequently $0 < R_U < p$. Since $hR_U = p + 4U \equiv 1 \pmod{4}$ and $h \equiv 3 \pmod{4}$, one has $R_U \equiv 3 \pmod{4}$. Thus a_U is integral and $p/4 < a_U < p/2$.

Prime by prime, $U \mid j^2$ implies $g_U^2/U \in \mathbb{Z}_{>0}$. The definitions give

$$h_U r_U = g_U, \quad h_U r_U^2 = U, \quad h_U r_U \lambda_U = j.$$

Moreover,

$$R_U j - U = \frac{R_U(h+1)}{4} - U = \frac{p + R_U}{4} = a_U.$$

It follows that $h_U r_U s_U = a_U$ and $r_U + s_U = R_U \lambda_U$. Since $\gcd(r_U, \lambda_U) = 1$, a common prime divisor of r_U, s_U would divide r_U, R_U , and hence $hR_U - 4U = p$. It cannot equal p , because $0 < R_U < p$. Thus $\gcd(r_U, s_U) = 1$. Finally

$$a_U + U = R_U j,$$

so $R_U \mid a_U + U$, and $U = h_U r_U^2 \mid a_U^2$. This proves the forward construction and $4h_U r_U \lambda_U - 1 = 4j - 1 = h$.

Conversely, take an original M state with cofactor $Q = h$, normalized as

$$a_M = h_M r_M s_M, \quad u_M = h_M r_M^2, \quad \lambda_M = \frac{r_M + s_M}{R_M}.$$

Then

$$j = \frac{h+1}{4} = h_M r_M \lambda_M, \quad \frac{j^2}{u_M} = h_M \lambda_M^2 \in \mathbb{Z}_{>0}.$$

Therefore $U = u_M \mid j^2$. Also

$$a_M + u_M = h_M r_M (r_M + s_M) = R_M j,$$

whence

$$p + 4U = 4(a_M + u_M) - R_M = R_M(4j - 1) = R_M h.$$

The primitive normalization gives $\gcd(j, U) = h_M r_M$, and (19) recovers h_M, r_M, λ_M, s_M exactly. Distinct U 's remain distinct as original middle divisors. This proves bijectivity. $\square$

Corollary 5 (Exact negative-square divisor slices). *If an original E state has $\alpha = -4c^2$ or $\beta = -4c^2$, then*

$$h \equiv 3 \pmod{4}, \quad \gcd(h, c) = 1,$$

and, with $j = (h+1)/4$,

$$\alpha = -4c^2 : \quad \mathcal{U}_{p,h} = \{U \mid j^2 : U \equiv c^2 \pmod{h}\}, \quad (21)$$

$$\beta = -4c^2 : \quad \mathcal{U}_{p,h} = \{U \mid j^2 : c^2 U \equiv j^2 \pmod{h}\}. \quad (22)$$

The words $U = c^2$ in alpha and $U = (j/c)^2$ in beta are available exactly when $c \mid j$. That divisibility is not necessary for the full fixed- Q fibre to be nonempty.

Proof. The congruence and coprimality assertions were proved in Theorem 3. In the alpha case, $p \equiv -4c^2 \pmod{h}$, so $h \mid p + 4U$ is equivalent to (21). In the beta case, $-4c^2 p \equiv 1 \pmod{h}$. Multiplication of $p + 4U \equiv 0 \pmod{h}$ by $4c^2$, together with $4j \equiv 1 \pmod{h}$, gives (22), and every step is reversible because $\gcd(4c^2, h) = 1$. $\square$

Define the finite residue set

$$\Delta_h = \{U \bmod h : U \mid j^2\} = \left\{ \prod_{\ell \parallel j} \ell^{f_\ell} \bmod h : 0 \leq f_\ell \leq 2e \right\}.$$

The exact emptiness criterion is

$$\boxed{\mathcal{U}_{p,h} = \emptyset \iff -p4^{-1} \pmod{h} \notin \Delta_h.} \quad (23)$$

Thus the fixed- Q question is a complete finite divisor-residue fibre, not merely the selected square test. Its other U 's can be nonsquares, so the square-grade conclusion of Theorem 3 remains specific to its selected words.

4.1. Composition with the diagonal and radius-eight returns. For a hard prime, let $\mathcal{E}_p$ be the complete original E-state set,

$$\mathcal{D}_p = \{e \in \mathcal{E}_p : R = D\},$$

$$\mathcal{F}_p = \{e \in \mathcal{E}_p : h \equiv 3 \pmod{4}, \mathcal{U}_{p,h} \neq \emptyset\}.$$

After applying the proved diagonal return and every fixed- Q return, the exact residual source is

$$\boxed{\mathcal{E}_p^{\text{res}} = \{e \in \mathcal{E}_p : R \neq D, [h \not\equiv 3 \pmod{4} \text{ or } \mathcal{U}_{p,h} = \emptyset]\}}. \quad (24)$$

Every state in (24) has

$$\boxed{\max(|\alpha|, |\beta|) \geq 9}. \quad (25)$$

Indeed, radius at most seven is the diagonal profile $\alpha = \beta = -1$. The sole nonsingular radius-eight profile is

$$(\alpha, \beta, h) = (8, -4, 11).$$

Here $j = 3$, and $U = 9 \in \mathcal{U}_{p,11}$; (19) gives

$$(h_U, r_U, s_U, \lambda_U) = \left(1, 3, \frac{p+3}{11}, 1\right),$$

$$a_U = \frac{3(p+3)}{11}, \quad R_U = \frac{p+36}{11}.$$

Thus the earlier radius-eight map is exactly the beta-square specialization of Theorem 4.

If a residual state nevertheless has a negative-square coordinate, it obeys the stronger finite exclusions

$$\alpha = -4c^2 \implies \nexists U \mid j^2 \text{ with } U \equiv c^2 \pmod{h},$$

$$\beta = -4c^2 \implies \nexists U \mid j^2 \text{ with } c^2 U \equiv j^2 \pmod{h}.$$

All earlier exact conditions remain: $h \mid \alpha\beta - 1$, $p \equiv \alpha \pmod{h}$, $p\beta \equiv 1 \pmod{h}$, $(h/p) = -1$, neither coordinate is zero or a positive square, and the exterior atlas bound is unchanged.

The overlaps are typed exactly. If one source has both negative-square coordinates, both descriptions parameterize the same full set $\mathcal{U}_{p,h}$; their selected targets coincide exactly when $c_\alpha c_\beta = j$. If δ is the squarefree part of $(R-1)/2$, the diagonal return coincides with a fixed- $Q = h$ target exactly when

$$\delta = h, \quad U = j.$$

Different U 's in $\mathcal{U}_{p,h}$ always produce different original middle states.

4.2. A strict hard-prime improvement. At

$$p = 41161, \quad h = 11, \quad j = 3, \quad c = 5,$$

the original E state

$$(a, R, u, h, r, s, \kappa, D, \alpha, \beta) = (10318, 111, 9678284, 11, 938, 1, 347829, 348767, -100, -348756)$$

has $\alpha = -4 \cdot 5^2$, although $5 \nmid 3$. The state equations are

$$a = 11 \cdot 938, \quad u = 11 \cdot 938^2, \quad 4a - p = 111, \quad 4u + 1 = 111 \cdot 348767,$$

$$pr + s = 111 \cdot 347829.$$

Nevertheless $U = 3 \mid 9$ and $p + 4U = 11 \cdot 3743$, so Theorem 4 gives

$$(a_U, R_U, U, h_U, r_U, s_U, \lambda_U) = (11226, 3743, 3, 3, 1, 3742, 1)$$

and

$$\boxed{\frac{4}{41161} = \frac{1}{11226} + \frac{1}{462073386} + \frac{1}{123483}}. \quad (26)$$

Primality is certified by the Lucas witness 22:

$$41160 = 2^3 \cdot 3 \cdot 5 \cdot 7^3, \quad 22^{41160} \equiv 1 \pmod{41161},$$

while the residues at exponents $41160/q$ for $q = 2, 3, 5, 7$ are

$$41160, \quad 40429, \quad 26424, \quad 13707,$$

and each residue minus one is coprime to 41161.

By contrast, at the prime $p = 3049$, the original E state

$$(a, R, u, h, r, s, \kappa, D, \alpha, \beta) = (806, 175, 20956, 31, 26, 1, 453, 479, -144, -448)$$

has ordered denominators $(806, 14043, 1113244782)$. Here $j = 8, c = 6$, and

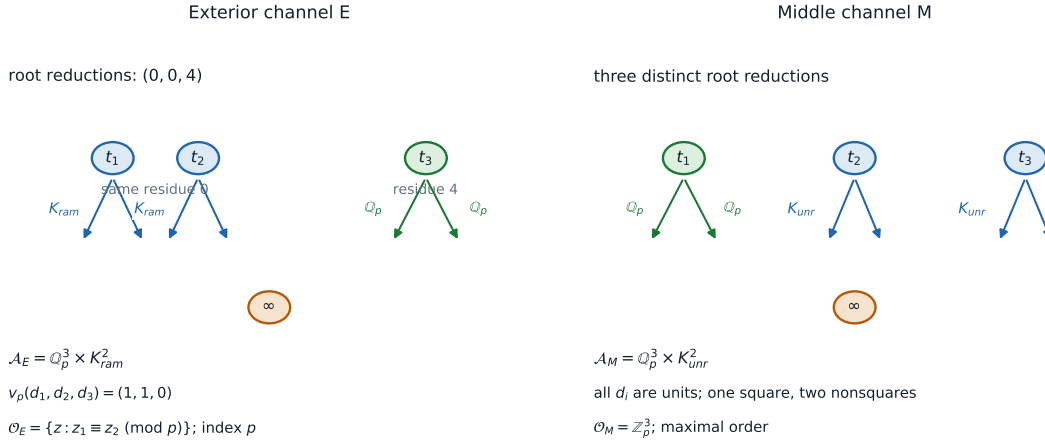
$$\{U \bmod 31 : U \mid 64\} = \{1, 2, 4, 8, 16\}, \quad c^2 = 36 \equiv 5 \pmod{31}.$$

Hence $\mathcal{U}_{3049,31}$ is empty. Primality is certified by the Lucas witness 11: $11^{3048} \equiv 1 \pmod{3049}$, while the residues at exponents $3048/q$ for $q = 2, 3, 127$ are 3048, 2516, 3041, each with predecessor coprime to 3049. A negative-square coordinate alone does not force a fixed- Q return.

5. PROOF BOUNDARY AND CERTIFICATE

The full source lists both inverse fibre parametrizations and all coefficient collisions. The standard-library certificate retains the original exponents, ordered denominators, all signed Fabel coordinates, Jacobian checks and Hensel lifts. A separate primitive-parameter implementation reproduces the complete bounded source and uses the original factor equations instead of the expanded polynomial. An inverse at an arbitrary cubic target does not prove an ES state: $(0, -4, 0, 1)$ has infinity inverse $(0, -4, 56i, -705)$ but its cubic has no rational root. Universal original target supply remains unresolved.

Prime-local reciprocal fibres retain the E/M arithmetic type



Each side has seven geometric affine branches and exactly three $\mathbb{Q}_p$ -points; the field and integral-order types differ.

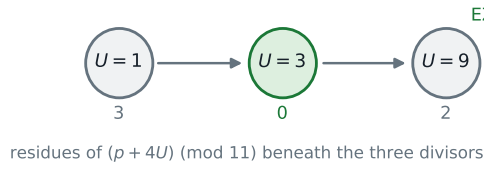
FIGURE 1. The two complete prime-local fibre algebras. Equal local-point counts do not identify the ramified exterior and unramified middle factors, and the exterior integral root order retains its exact index- p defect.

REFERENCES

- [1] Elsholtz, C., & Tao, T. (2013). Counting the number of solutions to the Erdős–Straus equation on unit fractions. *J. Aust. Math. Soc.*, 94, 50–105. [arXiv:1107.1010](#), §2.
- [2] Split-Zero programme. (2026). *The ES–Fable inverse correspondence in the original weighted conductor*. GF1–11, GF16–18, FC32–35. Zeta workbench revision 369f6ea9e0312bdf348ca9efd294735efc99c51d, source blob f777bb9f1434fa90f2f8bc762860048cc1d32bb4. The four-dimensional polynomial and inverse are antecedents, not newly claimed here.

Complete fixed- Q fibre at $p = 41161$, $h = 11$, $j = 3$

$$U \mid j^2 = 9$$



EZ28Q--S

Unique fixed- Q target ($U = 3$)

$$R_U = (p + 4U)/h = 3743, \quad A_U = (p + R_U)/4 = 11226$$

$$(h_U, r_U, s_U, \lambda_U) = (3, 1, 3742, 1)$$

$$4/41161 = 1/11226 + 1/462073386 + 1/123483$$

Why this strictly extends the selected square word

$$\alpha = -4c^2 = -100, \text{ so } c = 5 \text{ but } 5 \nmid j = 3.$$

Selected square word unavailable;
nonsquare $U = 3$ still returns.

General theorem: $\mathcal{V}_{p,h} = \{U > 0 : U \mid j^2, h \mid p + 4U\}$

$$\mathcal{V}_{p,h} \leftrightarrow \{ \text{all original M states with } Q = h \}$$

This fibre starts from an existing E state; it does not force occupancy.

FIGURE 2. At $p = 41161$, $h = 11$, $j = 3$, the complete divisor fibre contains the nonsquare word $U = 3$, which gives the displayed middle witness although the earlier selected square word is unavailable.